$PEPS

Peptide Perpetual Swaps Protocol

Technical Whitepaper v1.0

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Abstract

$PEPS introduces a novel decentralized perpetual swap protocol that enables leveraged trading on a basket of peptide biotech indices. By combining Automated Market Maker (AMM) mechanisms with oracle-based price feeds, the protocol achieves capital-efficient leverage of up to 5x while maintaining system solvency through dynamic liquidation mechanisms and funding rate adjustments. This paper presents the mathematical framework, smart contract architecture, and economic incentives that underpin the $PEPS ecosystem.

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1. Introduction

The peptide biotech sector represents a rapidly growing market with over $50 billion in annual revenue. Despite its significance, there exists no decentralized financial instrument that allows traders to gain leveraged exposure to peptide price movements. $PEPS fills this gap by introducing the first perpetual swap protocol specifically designed for peptide indices.

Traditional perpetual swap protocols focus on cryptocurrency and commodity markets. $PEPS extends this paradigm to biotech assets, creating a bridge between decentralized finance (DeFi) and pharmaceutical innovation. The protocol enables traders to express long or short views on peptide price indices with leverage ranging from 2x to 5x.

Key Innovations:

• Novel peptide price oracle aggregating multiple biotech data sources

• Capital-efficient leverage mechanism with dynamic collateral requirements

• Automated liquidation system ensuring protocol solvency

• Funding rate mechanism balancing long and short positions

• Gasless trading through meta-transactions

2. Protocol Architecture

The $PEPS protocol consists of four primary components: the Trading Contract, Oracle Contract, Staking Contract, and Governance Contract. Each component is implemented as an upgradeable smart contract on Ethereum Layer 2 solutions (Arbitrum and Base) to ensure low gas costs and high throughput.

Component	Function	Key Features
Trading Contract	Position management and settlement	Leverage control, PnL calculation, liquidations
Oracle Contract	Price feed aggregation	Multi-source validation, manipulation resistance
Staking Contract	Protocol insurance fund	Lock periods, yield generation, governance rights
Governance Contract	Parameter adjustment	Time-locked execution, community voting

3. Mathematical Framework

The $PEPS protocol employs sophisticated mathematical models to ensure fair pricing, efficient capital allocation, and system stability. This section details the core formulas governing position mechanics, liquidations, and funding rates.

3.1 Position Sizing and Leverage

When a trader opens a position, the protocol calculates the position size based on collateral and leverage. The fundamental relationship is:

$$\text{Position Size} = \text{Collateral} \times \text{Leverage}$$

The protocol supports leverage values $L \in \{2, 3, 4, 5\}$. The maximum position size is constrained by protocol limits to prevent excessive risk concentration. For a user depositing collateral C with leverage L at entry price P_{entry}, the position size S is:

$$S = (C \times L) / P_{entry}$$

3.2 Liquidation Mechanics

Liquidation occurs when the position's mark-to-market value falls below the maintenance margin. The liquidation price P_{liq} for a long position is calculated as:

$$P_{liq} = P_{entry} \times (1 - 1/L + \varepsilon)$$

where ε is a small safety buffer (typically 1%) to account for price volatility between oracle updates. For short positions, the formula becomes:

$$P_{liq} = P_{entry} \times (1 + 1/L - \varepsilon)$$

Example: A trader opens a long position with 1 ETH collateral at 3x leverage when the peptide index is at $1,000. Their position size is 3 ETH worth, or 0.003 peptide index units. The liquidation price is $1,000 × (1 - 1/3 + 0.01) = $677. If the index falls to $677, the position is liquidated.

3.3 Funding Rate Mechanism

To keep perpetual swap prices anchored to the oracle price, the protocol implements a funding rate mechanism. The funding rate r_f is calculated every 8 hours based on the imbalance between long and short open interest:

$$r_f = k \times (OI_{long} - OI_{short}) / (OI_{long} + OI_{short})$$

where k is a calibration constant (typically 0.01), OI_{long} is the total long open interest, and OI_{short} is the total short open interest. When longs dominate, $r_f > 0$, and long positions pay shorts. When shorts dominate, $r_f < 0$, and shorts pay longs.

The funding payment F for a position of size S over time period Δt is:

$$F = S \times P_{mark} \times r_f \times \Delta t / T$$

where T is the funding period (8 hours) and P_{mark} is the current mark price.

4. Oracle Design

The $PEPS oracle aggregates peptide pricing data from multiple sources to create a robust, manipulation-resistant price feed. The Peptide Price Index (PPI) is a weighted basket of leading peptide compounds traded on pharmaceutical markets.

4.1 Peptide Price Index Composition

Peptide	Weight	Category
GLP-1 Agonists	35%	Metabolic
BPC-157	25%	Regenerative
TB-500	20%	Recovery
MOTS-c	10%	Mitochondrial
Others	10%	Mixed

4.2 Data Sources and Aggregation

The oracle employs a median-of-medians algorithm to resist outliers and manipulation attempts. Price data is collected from N independent data providers ($N \geq 5$). For each peptide, the oracle:

1. Collects prices $p_1, p_2, ..., p_N$ from each provider
2. Removes the highest and lowest values
3. Calculates the median of remaining values
4. Computes the weighted average across all peptides in the index

The final index price P_{index} is:

$$P_{index} = \Sigma (w_i \times \text{median}(p_{i,1}, ..., p_{i,N-2}))$$

where w_i is the weight of peptide i in the index.

5. Smart Contract Architecture

The protocol is implemented across four primary smart contracts, each designed for upgradeability and gas efficiency. All contracts follow the OpenZeppelin upgradeable contract patterns and include comprehensive access control mechanisms.

Trading Contract

Key Functions:
• **openPosition(isLong, collateral, leverage)**: Opens a new leveraged position
• **closePosition(positionId)**: Closes an existing position and settles PnL
• **liquidate(positionId)**: Liquidates undercollateralized positions
• **updateFunding()**: Recalculates and distributes funding payments

The contract maintains a mapping of position IDs to Position structs containing:

```
struct Position { address trader; bool isLong; uint256 size;
  uint256 collateral; uint256 leverage; uint256 entryPrice;
  uint256 liquidationPrice; uint256 lastFundingUpdate; bool
                                isOpen; }
```

6. Economic Model

6.1 Fee Structure

The protocol generates revenue through trading fees and distributes them to $PEPS stakers. The fee structure is designed to incentivize long-term staking while maintaining competitive trading costs.

Fee Type	Rate	Distribution
Opening Fee	0.10%	70% to stakers, 30% to insurance fund
Closing Fee	0.10%	70% to stakers, 30% to insurance fund
Liquidation Fee	2.00%	50% to liquidator, 50% to insurance fund
Funding Rate	Variable	Between long and short traders

6.2 $PEPS Tokenomics

The $PEPS token serves as the governance and utility token of the protocol. Token holders can stake their tokens to earn protocol fees and participate in governance decisions.

Total Supply: 1,000,000,000 PEPS
Initial Circulating Supply: 200,000,000 PEPS (20%)

Token Distribution:
• 30% - Community rewards and liquidity mining
• 25% - Team and advisors (4-year vesting)
• 20% - Public sale
• 15% - Development fund
• 10% - Insurance fund

Staking Rewards Structure

Stakers earn rewards proportional to their lock period. The Annual Percentage Yield (APY) is calculated based on the total value locked (TVL) and protocol fee generation:

```
APY = (Annual Protocol Fees × 0.7) / TVL
```

Lock Period	Multiplier	Base APY
30 days	1.0x	12.5%
90 days	1.5x	18.75%
180 days	2.0x	25.0%
365 days	2.8x	35.0%

7. Risk Management

The protocol implements multiple risk management mechanisms to ensure long-term sustainability and protect both traders and liquidity providers.

Insurance Fund

The insurance fund acts as a backstop for liquidations that occur at prices worse than the liquidation price. The fund is capitalized through:

- 30% of all trading fees
- 50% of liquidation fees
- 10% of initial token supply

When a position is liquidated at a loss exceeding its collateral, the insurance fund covers the deficit. If the insurance fund is depleted, the protocol enters a socialized loss mode where losses are distributed among profitable traders proportionally.

Position Limits

To prevent excessive risk concentration, the protocol enforces position limits:

- Maximum position size: 10% of total protocol open interest
- Maximum leverage: 5x
- Maximum open positions per address: 20
- Minimum position size: $100 equivalent

8. Governance

$PEPS employs a decentralized autonomous organization (DAO) structure for protocol governance. Token holders can propose and vote on protocol changes including:

- Fee structure adjustments
- Maximum leverage modifications
- Oracle parameter updates
- Insurance fund allocation
- Protocol upgrades

Voting Power: 1 staked PEPS = 1 vote. Longer lock periods receive voting power multipliers matching their reward multipliers (up to 2.8x for 365-day locks).

Proposal Process:
1. Proposal submission (requires 100,000 PEPS)
2. Community discussion (7 days)
3. Voting period (7 days)
4. Time-lock execution (2 days)
5. Implementation

Quorum: 10% of total staked supply
Approval Threshold: 66.7% majority

9. Roadmap

Phase	Timeline	Milestones
Phase 1: Foundation	Q2 2026	• Smart contract deployment on Arbitrum • Basic trading interface • Single peptide index (GLP-1)
Phase 2: Expansion	Q3 2026	• Multi-peptide index launch • Mobile app release • Cross-chain bridge (Base)
Phase 3: Advanced Features	Q4 2026	• Limit orders and stop-losses • Portfolio margin system • Options on perpetuals
Phase 4: Maturity	Q1 2027	• Full DAO governance transition • Institutional API • Real-world asset integration

10. Conclusion

$PEPS represents a significant innovation in decentralized derivatives trading by bringing leveraged exposure to the peptide biotech sector. Through carefully designed mathematical frameworks, robust oracle infrastructure, and comprehensive risk management systems, the protocol achieves capital efficiency while maintaining security and fairness.

The protocol's unique combination of features positions it to capture significant market share in the emerging biotech DeFi sector. As the first mover in peptide perpetual swaps, $PEPS has the opportunity to establish the standard for biotech asset derivatives.

The roadmap demonstrates a commitment to continuous innovation and community-driven development. By progressively decentralizing governance and expanding feature sets, $PEPS aims to become the premier platform for leveraged peptide trading.

Key Takeaways:
• Novel application of perpetual swap mechanics to biotech assets
• Mathematically sound leverage and liquidation systems
• Robust oracle design resistant to manipulation
• Sustainable tokenomics rewarding long-term participation
• Clear governance framework for protocol evolution

The future of biotech finance is decentralized, and $PEPS is leading the way.